MILITARY SPECIFICATION

SPACER FOR CHARGE, PROPELLING, 175MM M124 (CELLULAR POLYURETHANE)

This specification is approved for use by Picatinny Arsenal Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a cellular polyurethane spacer for use with M124 Propelling Charge.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

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DRAWINGS

ARMY

9295042	- Packing & Marking for Box, Wirebound, Packing Ammo for Spacer for Charge Propelling 175mm Gun M124
9295043	- Box Wirebound
9295044	- Packing & Marking for Box, Fiberboard, Packing Ammo, for Spacer for Charge Propelling, 175mm Gun M124
9295045	- Box Fiberboard
9295046	- Bag Barrier
9298769	- Spacer, Charge Propelling

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Materials.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Spacer.-The Spacer shall comply with all requirements specified on Dwg. 9298769 and with all requirements specified in applicable specifications.

3.3 Density.-The density shall be 13.3 plus or minus 2.5 pounds per cubic foot throughout the length of the spacer when tested as specified in 4.5 .1.

3.4 Burning Characteristics.-The Polyurethane material shall start to disintegrate within 2 minutes and shall be completely consumed in 8 minutes minimum and 15 minutes maximum when tested as specified in 4.5.2.

3.5 Workmanship.-All parts shall be free of dirt, grease, oil and other foreign material except that traces of mold release agent will be permitted. There shall be no obvious structural deficiencies such as cracks.

3.6 First article inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of a minimum of 5 spacers from every mold cavity to be used.

4.3.2 Inspection to be performed.-Spacer will be subjected by the Government to any or all of the examinations or tests specified in this specification and any or all requirements of he applicable drawings. In addition, a 2 inch drop of a 150 pound weight to impact the spacer in it's axial orientation is to be conducted on one sample from each mold to determine adequacy of compressive strength. The spacer shall withstand collapse (i.e. a decrease in length of six inches or more).

4.3.3 Rejection.-See MIL-A-48078.

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II except that continuous sampling plans in accordance to MIL-STD-1235 may be used if approved by the Government.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Spacer				0208769
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length minimum		0.40%	3.2	Gage
102	Damage to the extent that the part is rendered ineffective for subsequent required use		0.40%	3.2	Visual
103	Density	5		3.3	4.5.1
104	Burning characteristics	10		3.4	4.5.2
Minor					
201	Wall thickness		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.5	Visual
203	Width at 2 locations 90° apart		0.65%	3.2	Gage

NOTES:

SARPA-QA FORM 2567 FEB 74 4910

MIL-S-63060 (PA)

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9295045
4.4.2.2	Fiberboard Box, Prior to Sealing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Any packing material missing		0.65%	3.2	Visual
202	Spacer missing		0.65%	3.2	Visual

NOTES

SARPA-QA FORM 2567 FEB 74 4010

MIL-S-63060 (PA)

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9295045
4.4.2.3	Fiberboard Box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Sealing strip torn, incomplete, badly wrinkled or otherwise fails to seal carton		0.65%	3.2	Visual
202	Carton torn, cut or punctured		0.65%	3.2	Visual
203	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
204	Contents loose		0.65%	3.2	Manual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Sealed Barrier Bag				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag punctured, torn or cut		0.40%	3.2	Visual
102	Seal mislocated or incomplete		0.40%	3.2	Visual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9295043
4.4.2.5	Sealed Packing Box				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box broken to extent that contents are exposed or liable to become exposed		0.40%	3.2	Visual
102	Strapping hardware missing, broken or loose		0.40%	3.2	Visual-Manual
Minor					
201	Strapping hardware improperly engaged or mis-located		0.65%	3.2	Visual
202	Marking Missing, misleading or unidentifiable		0.65%	3.2	Visual

NOTES:

SARPA-QA FORM 2567 FEB 74 4810

MIL-S-63060 (PA)

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4.4.3 Testing.-Testing is described in the Quality Conformance Inspection Tables.

4.4.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed in this specification is identified, either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the (First Article or) Quality Conformance Inspection Tables herein. The contractor shall submit inspection equipment designs to the Government for approval in accordance with the terms of the contract. See 6.3.

4.5 Test Method & Procedures

4.5.1 Density.-A specimen, between two and three inches in length, shall be cut from each end and one from the center (three specimens from each spacer). A section of spacer 2 1/2 plus or minus 1/2 inch in length shall be weighed in air and then weighed in distilled water with a weight attached. The weight used shall also be weighed separately in distilled water. The temperature of the water shall be 23 plus or minus 1 degree Centigrade and a small amount of wetting agent may be added. If any specimen fails to comply with the requirement of 3.3 the lot shall be rejected. The specific gravity shall be calculated as follows:

$$\text{Specific gravity} = \frac{A}{A - (B-C)}$$

where:

A = weight of specimen in air
B = weight of specimen plus weight in water
C = weight of weight in water

The density shall be calculated as follows:

Density = specific gravity x 62.43

4.5.2 Burning characteristics.-Each specimen shall weigh 5 plus or minus 1 gram. The specimen shall be placed in muffle furnace maintained at a temperature of 1000 degrees plus or minus 50 degrees Centigrade. Observations shall be made at one minute intervals until the specimen starts to disintegrate and every two minutes thereafter until the sample is completely consumed. If any specimen fails to comply with the requirements of 3.4 the lot shall be rejected.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging

5.1.1 Level C.-The spacers shall be preserved, packaged and marked in accordance with Dwg. 9295045 and 9295044.

5.2 Packing

5.2.1 Level C.-The box shall be packed and marked in accordance with Dwg. 9295043 and 9295042.

6. NOTES

6.1 Intended Use.-The spacer covered by this specification is intended for use with 175MM M124 Propelling Charge.

6.2 Ordering Data.-See MIL-A-48078 (PA).

6.3 Submission of Inspection Equipment Designs for Approval.-See MIL-A-48078. Submit equipment designs, as required to:

Commander
Picatinny Arsenal
ATTN: SARPA-QA-T-T
Dover, New Jersey 07801

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1320-A325

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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DATE

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Width of rotating band

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COMMANDER
Picatinny Arsenal
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Dover, New Jersey 07801

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